

ANH MAI VU

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SUMMARY

PhD candidate in Computer Engineering at the University of Houston seeking an **AI/ML Research Internship**. My research focuses on trustworthy AI, explainable AI, computer vision, multimodal learning, medical image analysis, visual grounding, world models in medicine, and human-centered AI.

EDUCATION

PhD in Computer Engineering

Aug 2024 – Present

University of Houston, USA

Topic: World Models in Medicine; Computational Pathology, Trustworthy AI, Gaze-Guided Explainability

MSc in Mathematical Foundations of Computer Science (Data Science)

Vietnam National University – Ho Chi Minh City, Vietnam GPA 8.97/10

Bachelor of Business Administration

University of Economics and Law, Ho Chi Minh City, Vietnam

EXPERIENCE

Graduate Research Assistant

Aug 2024 – Present

Department of Computer Engineering, University of Houston, USA

Supervisor: Prof. Hien Van Nguyen

- Develop explainable AI and computer vision methods for whole-slide pathology image analysis using deep learning, weakly supervised learning, visual grounding, and model attribution.
- Build multimodal AI models that integrate medical images, clinical text, and eye-gaze data to align model reasoning with expert diagnostic behavior.
- Implement research pipelines in Python and PyTorch for image segmentation, feature attribution, prototype learning, and multimodal representation learning.
- Published first-author papers at WACV 2026, ISBI 2026, and MIDL 2026; contributed to MICCAI 2025, Scientific Reports, and Knowledge-Based Systems.

Research Assistant - Research Internship (NSTC Scholarship Program, Taiwan)

Oct 2023 – Jan 2024

National Cheng Kung University - BME Lab (Cancer Organoid Research)

Supervisor: Prof. Ting-Yuan Tu

- Applied deep learning and computer vision methods in a biomedical engineering lab to analyze functional behavior in human cardiac organoids, enabling real-time quantification of contraction dynamics from microscopy videos.

Senior Data Analyst, Machine Learning Focus

Jan 2019 – Oct 2022

Tiki Corporation, Vietnam

- Designed and deployed production recommendation systems using collaborative filtering, content-based models, and hybrid machine learning approaches.
- Built scalable end-to-end data and ML pipelines using Python, SQL, BigQuery, and Apache Airflow.
- Evaluated machine learning models and product changes through A/B testing, performance monitoring, and business metric analysis.
- Led data-driven modeling projects involving customer segmentation, clustering, predictive analytics, and automated reporting.

Data Analyst

Nov 2017 – Jan 2019

GHN Logistics, Vietnam

- Developed data warehouses, automated reporting systems, and analytical dashboards using Python, SQL, and Google Apps Script.

PUBLICATIONS

*Equal contribution

- *Contrastive Integrated Gradients: A Feature Attribution-Based Method for Explaining Whole Slide Image Classification.*
Anh Mai Vu, T.L. Vo, N.L.Q. Bui, N.N.L. Binh, A. Awasthi, H.Q. Vo, T.H. Nguyen, Z. Han, C. Mohan, H.V. Nguyen
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026
- *LPD: Learnable Prototypes with Diversity Regularization for Weakly Supervised Histopathology Segmentation.*
Anh Mai Vu*, Khang Le*, T.K.T. Vo, H. Thach, N.L.Q. Bui, T.H. Nguyen, M.H.N. Le, Z. Han, C. Mohan, H.V. Nguyen
IEEE International Symposium on Biomedical Imaging (ISBI), 2026
- *ConStruct: Structural Distillation of Foundation Models for Prototype-Based Weakly Supervised Histopathology Segmentation.*
Anh M. Vu*, Khang Le*, Ha Thach*, Trang T.K. Vo, Han H. Huynh, David Yang, Minh H.N. Le, Thanh-Huy Nguyen, Akash Awasthi, Chandra Mohan, Zhu Han, Hien Van Nguyen
Medical Imaging with Deep Learning (MIDL), 2026
- *Structural Chain of Thoughts for Radiology Education.*
A. Awasthi, B. Chung, A.M. Vu, S. Khan, N. Le, Z. Deng, R. Agrawal, C.C. Wu, H. Van Nguyen
Knowledge-Based Systems (Elsevier), 2025 (IF: 7.6, Q1)
- *Modeling radiologists' cognitive processes using a digital gaze twin to enhance radiology training.*
A. Awasthi, Anh Mai Vu, N. Le, Z. Deng, S. Maulik, R. Agrawal, C.C. Wu, H.V. Nguyen
Scientific Reports (Springer Nature), 2025 (IF:3.9, Q1)
- *MAARTA: Multi-Agent Adaptive Radiology Teaching Assistant.*
A. Awasthi, B.V. Chung, Anh M. Vu, N. Le, R. Agrawal, Z. Deng, C. Wu, H.V. Nguyen
International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2025
- *Scribble-Supervised Medical Image Segmentation with Dynamic Teacher Switching and Hierarchical Consistency.*
H.T. Nguyen, L.H. Cao, D.T. Chung, Anh Mai Vu, T.-M. Nguyen, M.H.N. Le, P.K. Huynh, U. Bagci
IEEE International Symposium on Biomedical Imaging (ISBI), 2026
- *IMACT-CXR: An Interactive Multi-Agent Conversational Tutoring System for Chest X-Ray Interpretation*
Tuan-Anh Le, Anh Mai Vu, David Yang, Akash Awasthi, Hien Van Nguyen
IEEE International Symposium on Biomedical Imaging (ISBI), 2026
- *QwenVLConnector: A Fast, Unified Medical VLM Chatbot for Fine-Grained Clinical Perception and Text Generation.*
T. Nguyen, T.H. Nguyen, G.M. Hoang, A.M. Vu, U. Bagci
MICCAI 2025 FLARE Challenge
- *Lightweight Dual-Task Framework for Semi-supervised Lesion Segmentation with Knowledge Distillation from SAM.*
X.-L. Huynh, H.-T. Pham, Anh Mai Vu, T.-M. Nguyen, T.Q.K. Bui, T.-B. Nguyen, Q. Nguyen, M.H.N. Le, P.K. Huynh
MICCAI Workshop on Deep Generative Models (DGM), 2025
- *Blurry-Consistency Segmentation Framework with Selective Stacking on Differential Interference Contrast 3D Breast Cancer Spheroid.*
T.H. Nguyen, T.K.N. Ngo, M.A. Vu, T.Y. Tu
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2024
- *Correlation visualization under missing values: a comparison between imputation and direct parameter estimation methods.*
M.A. Vu*, N.H. Pham*, K.L. Vo*, T. Nguyen, M.A. Riegler, P. Halvorsen, B.T. Nguyen
International Conference on Multimedia Modeling (MMM), 2024
- *DualProtoSeg: Simple and Efficient Design with Text- and Image-Guided Prototype Learning for Weakly Supervised Histopathology Image Segmentation.*
Anh Mai Vu*, Khang P. Le*, Trang T.K. Vo*, Ha Thach, Huy Hung Nguyen, David Yang, Han H. Huynh, Quynh Nguyen, Tuan M. Pham, Tuan-Anh Le, Minh H.N. Le, Thanh-Huy Nguyen, Akash Awasthi, Chandra Mohan, Zhu Han, Hien Van Nguyen
Under Review, 2026

SERVICE

Technical Chair, Emerging LLM/LMM Applications in Medical Imaging (ELAMI) Workshop, MICCAI 2026

2026

AWARDS & SELECTED MENTIONS

- MICCAI FLARE Challenge 2025 Finalist with Honorable Award. [Cert]
- AI Challenge 2023, National Video Retrieval Competition, Finalist.
- Featured by **Tuoi Tre**, a leading Vietnamese newspaper, in a story about my academic journey and research interests in AI for medicine, Oct 2025.

SKILLS

Frameworks: PyTorch, TensorFlow

Programming Languages: Python, Bash

Libraries and Tools: NumPy, Pandas, scikit-learn, Matplotlib, Docker, Git, SLURM

Areas of Expertise: Computer Vision, Deep Learning, Self-Supervised Learning, Weakly Supervised Learning, Parameter-Efficient Fine-Tuning, Large Language Models, Multimodal AI, Explainable AI, Medical Image Analysis

Data and Infrastructure: BigQuery, Apache Airflow, Google Apps Script, Git, Linux